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Name:

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Pymble Ladies' College

**HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION
2014**

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes.
- Working time – 2 hours.
- Write using pencil for Questions 1-10.
- Write using black or blue pen for Questions 11-14. Black pen is preferred.
- Board approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- In Questions 11-14, show relevant mathematical reasoning and/or calculations.

Total Marks – 70

Section I Pages 1-4

10 marks

- Attempt all Questions 1-10
- Allow about 15 mins for this section

Section II Pages 5-11

60 marks

- Attempt Questions 11-14
- Allow about 1 hour 45 minutes for this section

Mark	/70
Highest Mark	/70
Rank	

Section I

10 marks

Attempt Questions 1-10

Allow about 15 minutes for this section.

Use the multiple choice answer sheet for Questions 1-10.

- 1 The roots of the equation $x^3 - 5x^2 + 4 = 0$ are α, β and γ .

The value of $\alpha + \beta + \gamma$ and the value of $\alpha\beta\gamma$ are respectively.

- (A) 5 and 4
- (B) 5 and -4
- (C) -5 and 4
- (D) -5 and -4

- 2 Evaluate $\sin^{-1}\left(\sin\frac{4\pi}{3}\right)$.

- (A) $\frac{4\pi}{3}$
- (B) $\frac{\pi}{3}$
- (C) $\frac{-2\pi}{3}$
- (D) $\frac{-\pi}{3}$

- 3 When the polynomial $P(x) = x^4 + ax + 2$ is divided by $x^2 + 1$ the remainder is $2x + 3$.

The value of a is

- (A) 1
- (B) 2
- (C) 0
- (D) 3

- 4 Given the points $A(7, 14)$ and $B(1, 2)$, C is a point on AB produced such that $AB : BC = 2 : 1$.

Find the coordinates of C .

- (A) $(-5, -10)$
- (B) $(-2, -4)$
- (C) $(3, 6)$
- (D) $(5, 10)$

- 5 Find $\int \frac{1}{\sqrt{1-3x^2}} dx$.

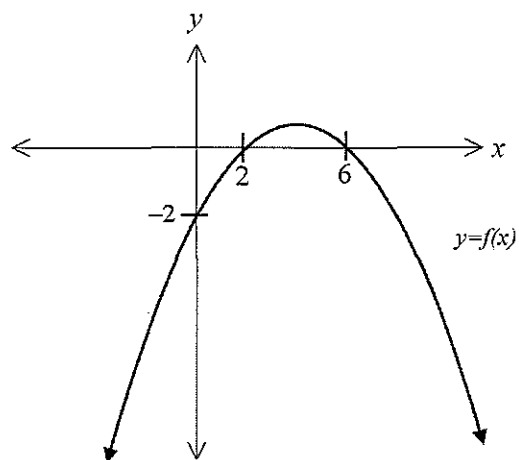
- (A) $3\sin^{-1}(3x) + C$
- (B) $\frac{1}{3}\sin^{-1}(3x) + C$
- (C) $\sqrt{3}\sin^{-1}(\sqrt{3}x) + C$
- (D) $\frac{1}{\sqrt{3}}\sin^{-1}(\sqrt{3}x) + C$

- 6 Evaluate $\int_0^{\frac{\pi}{6}} \sin^2 \theta d\theta$.

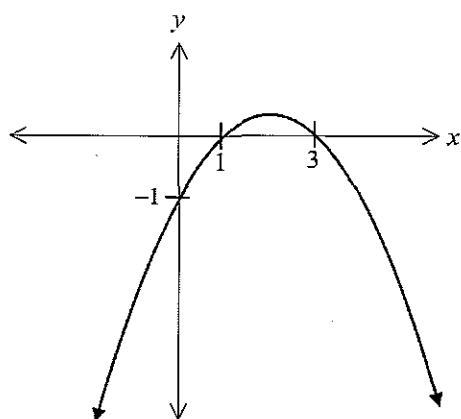
- (A) $\frac{\pi}{12} - \frac{\sqrt{3}}{8}$
- (B) $\frac{\pi}{6} - \frac{\sqrt{3}}{4}$
- (C) $\frac{1}{24}$
- (D) 1

- 7 The figure on the right shows the graph of $y = f(x)$.

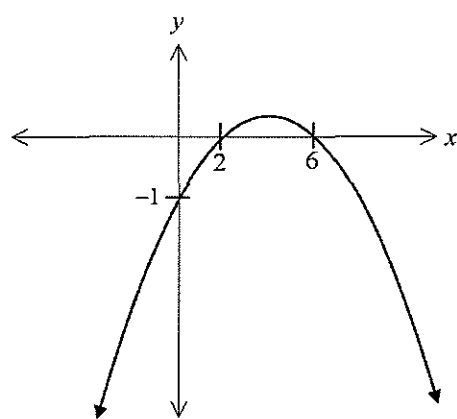
If $2f(x) = g(x)$, which of the following may represent the graph of $y = g(x)$?



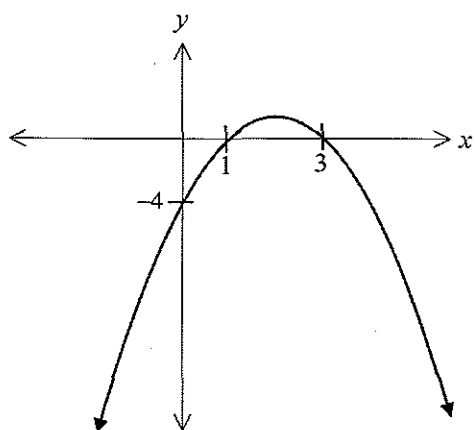
(A)



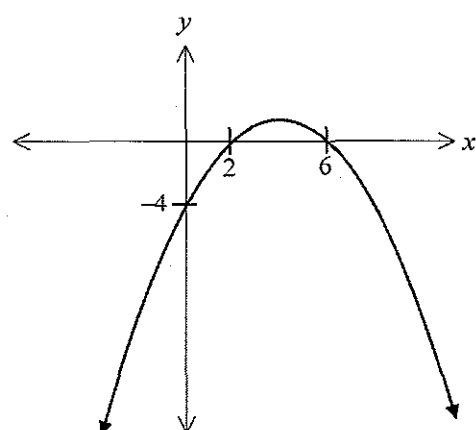
(B)



(C)



(D)



- 8 If $\int_{-a}^a f(x) dx = 0$, then which one of the following statements is false?
- (A) $f(x)$ is an odd function
- (B) $\int_0^a f(x) dx = \int_{-a}^0 f(-x) dx$
- (C) $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$
- (D) The area bounded by the curve $y = f(x)$, the x axis and the lines $x = a$ and $x = -a$ is twice the area bounded by the curve $y = f(x)$, the x axis and the lines $x = 0$ and $x = a$.

- 9 For $0^\circ \leq \theta \leq 90^\circ$, the least value of $\frac{30}{3\sin^2 \theta + 2\sin^2 (90^\circ - \theta)}$ is

- (A) 5
- (B) 6
- (C) 10
- (D) 15

- 10 Given n is an integer, the general solution of $\tan\left(2x + \frac{\pi}{4}\right) = \sqrt{3}$ is

- (A) $x = \frac{(12n+1)\pi}{24}$
- (B) $x = \frac{(3n+1)\pi}{6}$
- (C) $x = \frac{(12n-1)\pi}{24}$
- (D) $x = \frac{(6n+1)\pi}{6}$

Section II

60 marks

Attempt Questions 11-14

Allow about 1 hours and 45 minutes for this section.

Answer each question in a SEPARATE writing booklet. Extra booklets are available.

In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11. (15 marks). Use a **Separate Booklet**.

Marks

- (a) Given $f(x) = x^4 + x^2 - 80$.

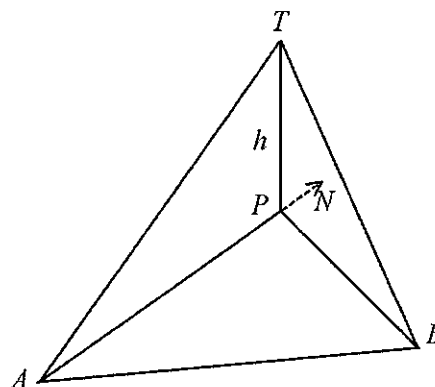
2

Assume there is a zero near $x = 3$. Use Newton's method once to find a better approximation to the root correct to 2 significant figures.

- (b) From a point A due south of a tower, TP , the angle of elevation of the top of the tower, T is 25° and from a point B due east of the tower is 32° .

The distance from A to B is 50 metres.

Let the height of tower TP be h metres.



- (i) Copy the diagram in your answer booklet and complete with all given information.

1

- (ii) Find an expression for PA in terms of h .

1

- (iii) Find the height of the tower, h , correct to 1 decimal place.

3

Question 11 continues on page 6.

(c) The function $f(x)$ is defined as $f(x) = \frac{3x-4}{x+2}$, where $x \neq -2$.

(i) Find an expression for $f^{-1}(x)$. 2

(ii) Write down the domain of $f^{-1}(x)$. 1

(d) Solve $\frac{4}{(x-1)^2} > 1$. 3

(e) Find $\int \frac{\ln x}{2x} dx$ using the substitution $u = \ln x$. 2

End of Question 11

(a) Find the term independent of x in the expansion of $\left(2x + \frac{1}{x^2}\right)^6$. **2**

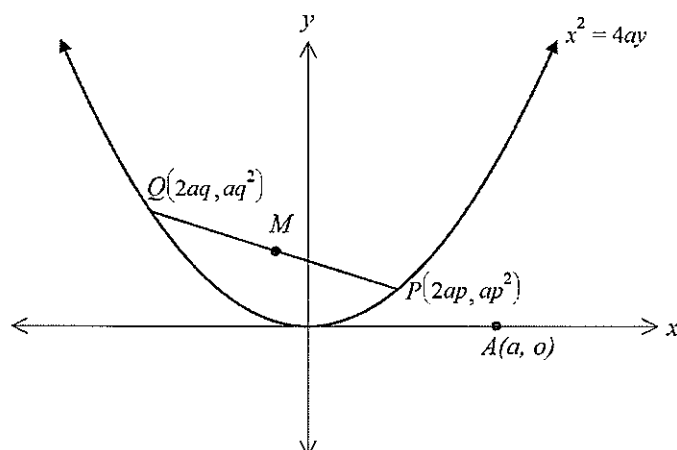
(b) (i) Show that $\tan x = \frac{\sin 2x}{1 + \cos 2x}$. **2**

(ii) Hence evaluate $\tan \frac{\pi}{12}$ in simplest form. **2**

(c) Prove by mathematical induction that $8^n - 3^n$ is divisible by 5, where n is a positive integer. **3**

Question 12 continues on page 8.

(d)



In the diagram above, the points $P(2ap, ap^2)$ and $Q(2aq, aq^2)$ lie on the parabola with equation $x^2 = 4ay$.

- (i) Write down the coordinates of the midpoint M of the chord PQ . 1
- (ii) Show that the equation of the chord PQ is $y = \frac{(p+q)x}{2} - apq$. 2
- (iii) Show that the condition for the chord PQ produced to pass through the point $A(a, 0)$ is $p + q = 2pq$. 1
- (iv) Find the cartesian equation of the locus of M , as the points P and Q move on the parabola subject to the constraint that PQ pass through $A(a, 0)$. 2

End of Question 12

- (a) Find the acute angle between the tangents on the curve $y = \tan^{-1} x$ at the points where $x = 0$ and $x = 1$. Answer correct to the nearest degree. 2

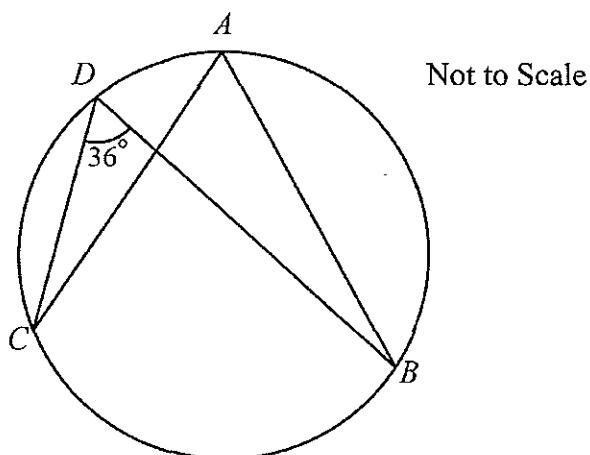
- (b) During a chemical reaction, the amount, R kg, of chemical formed at time t hours is modelled by the differential equation

$$\frac{dR}{dt} = 4 - \frac{R}{15}.$$

- (i) Show that $R = 60 - 50e^{-\frac{t}{15}}$ is a solution to $\frac{dR}{dt} = 4 - \frac{R}{15}$. 2

- (ii) How long will it take for 20 kg of the chemical to form? Give your answer correct to 2 significant figures. 2

- (c) In the figure below, BD is a diameter of the circle $ABCD$. If $AB = AC$ and $\angle BDC = 36^\circ$, find $\angle ABD$. 3



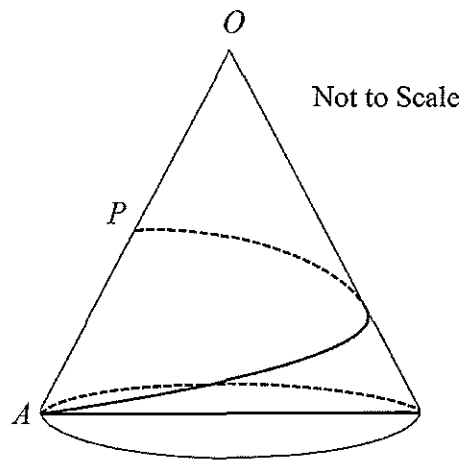
Question 13 continues on page 10.

- (d) A thin sheet of smooth metal is in the shape of a sector of a circle with OA , OB as bounding radii each of length 10 cm, and the angle AOB is 60° .

- (i) Find the length of the arc AB .

1

- (ii) The sheet is now bent to form a right circular cone by welding the radii OA and OB together (and inserting a circular disc to close in the cone at the base).



- (α) Find the volume of the cone in terms of π .

3

(Note: The volume of a right circular cone is, $\frac{1}{3}\pi r^2 h$.)

- (β) On the surface of this cone a thin string is pulled tight starting with one end fixed at the point A and passing once round the cone to the other end P which is at the midpoint of OA (as shown in diagram).

2

Find the exact length of this string.

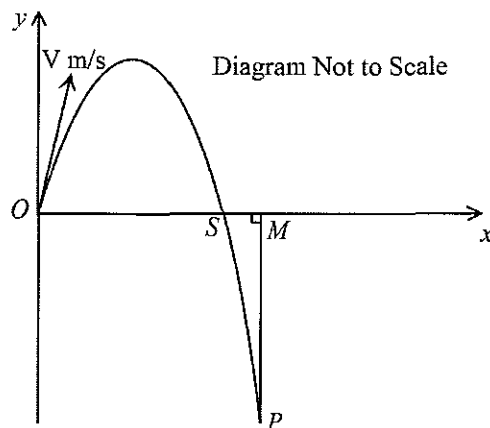
End of Question 13

- (a) Solve $\sin x - 3\cos x = 3$ for $0^\circ \leq x \leq 360^\circ$. 3

- (b) A projectile is fired from a point O with initial speed of V m/s at an angle of elevation θ . If x and y are the horizontal and vertical displacements of the projectile in metres from O at time t seconds later then

$$x = Vt \cos \theta \text{ and } y = Vt \sin \theta - \frac{1}{2}gt^2 \text{ where } g \text{ m/s}^2 \text{ is the acceleration due to gravity.}$$

The projectile falls to a point P below the level of O such that $PM = OM$.



- (i) Prove that the time taken to reach P is $2V \frac{(\sin \theta + \cos \theta)}{g}$ seconds. 1
- (ii) Show that the distance OM is 2

$$\frac{V^2}{g}(\sin 2\theta + \cos 2\theta + 1) \text{ metres.}$$
- (iii) If $OS = r$, $OM = \frac{4r}{3}$ and $r > 0$, prove that $\sin 2\theta - 3\cos 2\theta = 3$. 3
- (iv) Hence, by using Question 14 part (a), find the value of θ . 2
- (v) Find an expression for the horizontal and vertical components of the velocity. 1
- (vi) If the magnitude of the velocity of the projectile at P is kV m/s, find the exact value of k . 3

End of Paper

1. B

2. D

3. B

4. B

5. D

6. A

7. D

8. C

9. C

10. A

Q 11

$$a) f(x) = x^4 + x^2 - 80$$

$$f'(x) = 4x^3 + 2x$$

$$x_2 = 3 - \frac{f(3)}{f'(3)}$$

$$= 3 - \frac{3^4 + 3^2 - 80}{4 \times 3^3 + 2 \times 3}$$

$$= 2.912 \dots$$

$$= 2.9$$

(2)

b) ii) In $\triangle APT$;

$$\tan 25^\circ = \frac{h}{PA}$$

$$PA = h \cot 25^\circ \quad (1)$$

iii) In $\triangle BPT$;

$$\tan 32^\circ = \frac{h}{PB}$$

$$PB = h \cot 32^\circ \quad A$$

In $\triangle APB$;

$$50^2 = (h \cot 25^\circ)^2 + (h \cot 32^\circ)^2$$

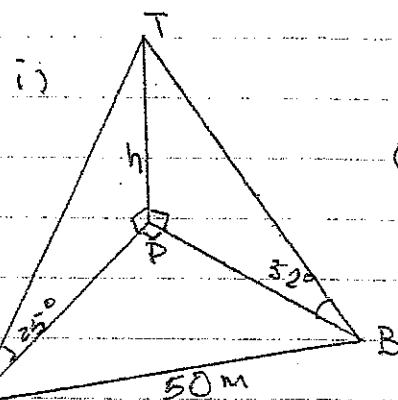
$$h^2 = \frac{50^2}{\cot^2 25^\circ + \cot^2 32^\circ}$$

$$= 349.16 \dots$$

$$h = 18.68 \dots \quad (h \text{ is a height of tower})$$

$$= 18.7 \text{ m}$$

(3)



$$c) f(x) = \frac{3x-4}{x+2}$$

$$ii) x = \frac{3y-4}{y+2}$$

$$xy + 2x = 3y - 4$$

$$xy - 3y = -4 - 2x$$

$$3y - xy = 2x + 4$$

$$y = \frac{2x+4}{3-x}$$

$$f^{-1}(x) = \frac{2x+4}{3-x}$$

(2)

iii) Domain: all real x ; $x \neq 3$

(1)

$$d) \frac{4}{(x-1)^2} > 1 \quad ; \quad x \neq 1$$

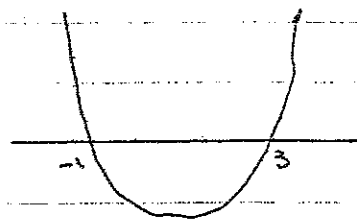
$$4 > x^2 - 2x + 1$$

$$x^2 - 2x - 3 < 0$$

$$(x-3)(x+1) < 0$$

$$-1 < x < 3$$

$\therefore -1 < x < 3$ except $x = 1$.



(3)

$$e) \int \frac{\ln x}{2x} dx \quad ; \quad u = \ln x$$

$$= \frac{1}{2} \int \ln x \cdot \frac{1}{x} dx$$

$$du = \frac{1}{x} dx$$

$$= \frac{1}{2} \int u du$$

$$= \frac{1}{2} \left(\frac{1}{2} u^2 \right) + C$$

$$= \frac{1}{4} u^2 + C$$

$$= \frac{1}{4} (\ln x)^2 + C$$

(2)

Q 12

$$a) (2x + \frac{1}{x^2})^6$$

$$= \sum_{r=0}^6 {}^6C_r (2x)^r (x^{-2})^{6-r}$$

Term independent of x

$$= {}^6C_4 (2x)^4 (x^{-2})^2$$

$$= {}^6C_4 \cdot 2^4$$

$$= 240$$

(2)

$$b) i) RHS = \frac{\sin 2x}{1 + \cos 2x}$$

$$= \frac{2 \sin x \cos x}{1 + 2 \cos^2 x - 1}$$

$$= \frac{2 \sin x \cos x}{2 \cos^2 x}$$

$$= \frac{\sin x}{\cos x}$$

$$= \tan x$$

$$= LHS$$

(2)

$$ii) \tan \frac{\pi}{12}$$

$$= \frac{\sin \frac{\pi}{6}}{1 + \cos \frac{\pi}{6}}$$

$$= \frac{\frac{1}{2}}{1 + \frac{\sqrt{3}}{2}}$$

$$= \frac{\frac{1}{2}}{\frac{\sqrt{3} + 2}{2}}$$

$$= \frac{1}{\sqrt{3} + 2}$$

$$= 2 - \sqrt{3}$$

(2)

c) Let the statement be $8^n - 3^n = 5P$ where P is an integer.

When $n = 1$;
 $8^1 - 3^1 = 8 - 3$

$= 5$ which is divisible by 5.

Assume the statement is true for $n = k$;

i.e. $8^k - 3^k = 5P \Rightarrow 8^k = 5P + 3^k$ —

Prove that the statement is true for $n = k + 1$;

i.e. prove that $8^{k+1} - 3^{k+1} = 5Q$ where Q is an integer.

LHS = $8^{k+1} - 3^{k+1}$

$= 8(8^k) - 3^{k+1}$

$= 8(5P + 3^k) - 3(3^k)$; from *

$= 5 \cdot 8P + 8 \cdot 3^k - 3 \cdot 3^k$

$= 5 \cdot 8P + 5 \cdot 3^k$

$= 5(8P + 3^k)$; since P is an integer, $8P$ is an integer, and since k is a positive integer, 3^k is an integer.

$= 5Q$; $Q = 8P + 3^k$ is also an integer.

Since the statement is true for $n = 1$,

assume true for $n = k$ and proved true for $n = k$.

So the statement is true for $n = 1 + 1 = 2$,

$n = 2 + 1 = 3, \dots$, $\therefore$ the statement is true for all positive integers of n .

1. for intro, * and conclusion

1. for case $n = 1$

1. for correct steps showing

$8^{k+1} - 3^{k+1} = 5Q$, Q an integer

3

ds is Midpoint M of PQ

$$= \left(\frac{2ap + 2aq}{2}, \frac{ap^2 + aq^2}{2} \right)$$

$$= (ap + aq, \frac{ap^2 + aq^2}{2}) \quad (1)$$

$$ii) m_{PQ} = \frac{ap^2 - aq^2}{2ap - 2aq}$$

$$= \frac{a(p+q)(p-q)}{2a(p-q)}$$

$$= \frac{1}{2}(p+q)$$

Equation of chord PQ is

$$\frac{p+q}{2} = \frac{y - ap^2}{x - 2ap}$$

$$\left(\frac{p+q}{2} \right) x - 2ap \left(\frac{p+q}{2} \right) = y - ap^2$$

$$\frac{(p+q)x}{2} - (ap^2 + apq) + ap^2 = y$$

$$y = \frac{(p+q)x}{2} - ap^2 - apq + ap^2$$

$$y = \frac{(p+q)x}{2} - apq \quad (2)$$

iii) If chord PQ passes through A(a, 0),

$$\text{then } 0 = \frac{1}{2}a(p+q) - apq \quad (1)$$

$$\frac{1}{2}a(p+q) = apq$$

$$\frac{1}{2}(p+q) = pq$$

$$p+q = 2pq$$

$$iv) x_M = ap + aq$$

$$x_M = a(p+q)$$

$$x_M = 2apq \quad ; \quad p+q = 2pq$$

$$y_M = \frac{1}{2}a(p^2 + q^2)$$

$$= \frac{1}{2}a[(p+q)^2 - 2pq]$$

$$= \frac{1}{2}a(p+q)^2 - apq$$

$$= \frac{1}{2a}[a^2(p+q)^2] - apq$$

$$= \frac{1}{2a}(x_M)^2 - \frac{1}{2}x_M$$

$$2ay = x^2 - ax$$

$$x^2 - ax - 2ay = 0 \quad (2)$$

Q 13

a) $y = \tan^{-1} x$

$$y' = \frac{1}{1+x^2}$$

$$m_1 = \frac{1}{1+0^2} = 1 \quad ; \quad m \text{ of tangent at } x=0$$

$$m_2 = \frac{1}{1+1^2} = \frac{1}{2} \quad ; \quad m \text{ of tangent at } x=1$$

$$\tan \theta = \frac{1 - \frac{1}{2}}{1 + (1)(\frac{1}{2})}$$

$$\tan \theta = \frac{1}{3}$$

The acute angle, $\theta = 18.43 \dots$

$$= 18^\circ$$

b) i) $R = 60 - 50e^{-\frac{t}{15}}$

$$\Rightarrow 50e^{-\frac{t}{15}} = 60 - R$$

$$\frac{dR}{dt} = 0 - 50\left(\frac{-1}{15}e^{-\frac{t}{15}}\right)$$

$$= \frac{1}{15}(50e^{-\frac{t}{15}})$$

$$= \frac{1}{15}(60 - R)$$

$$= 4 - \frac{R}{15} \quad (2)$$

ii) $20 = 60 - 50e^{-\frac{t}{15}}$

$$-40 = -50e^{-\frac{t}{15}}$$

$$\frac{-t}{15} = \ln \frac{4}{5}$$

$$t = -15 \ln \frac{4}{5}$$

$$= 3.347 \dots$$

$$\approx 3.3 \text{ hours} \quad (2)$$

c) $\angle BAC = \angle BDC$ (angles at the circumference standing on the same chord BC)
 $= 36^\circ$

$\angle ACB = \angle ABC$ (angles opposite equal sides AB and AC in isosceles $\triangle ABC$)

$\angle ACB + \angle ABC + \angle BAC = 180^\circ$ (angles sum of $\triangle ABC$)

$2\angle ACB + 36^\circ = 180^\circ$

$\angle ACB = 72^\circ$

$\angle DCA = \angle DBA$ (angles at the circumference standing on the same arc AD)

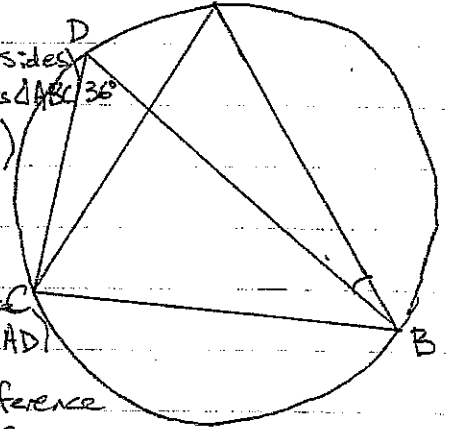
$\angle DCB = 90^\circ$ (angles at the circumference in a semicircle, BD is the diameter)

$\angle DCA + \angle ACB = \angle DCB$ (adjacent angles)

$\angle DBA + 72^\circ = 90^\circ$

$\angle DBA = 90^\circ - 72^\circ$

$= 18^\circ$



(3)

$$\therefore \angle AOB = 60^\circ = \frac{\pi}{3}$$

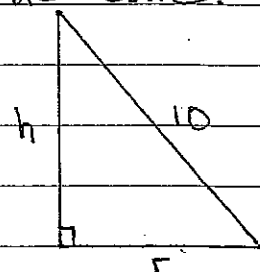
$$\begin{aligned} \text{Length of arc AB} &= 10 \times \frac{\pi}{3} \\ &= \frac{10\pi}{3} \text{ cm} \end{aligned} \quad (1)$$

ii) a) Let r be the radius of the cone and h be the height of the cone.

$$2\pi r = \frac{10\pi}{3}$$

$$r = \frac{5}{3} \text{ cm}$$

$$\begin{aligned} h &= \sqrt{10^2 - \left(\frac{5}{3}\right)^2} \\ &= \frac{5\sqrt{35}}{3} \text{ cm} \end{aligned}$$

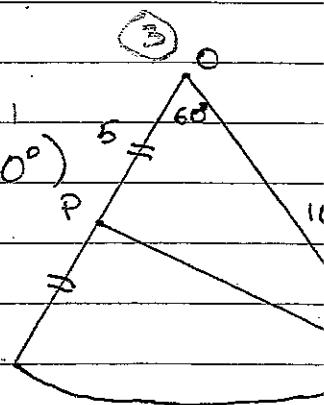


$$\begin{aligned} \text{Volume of cone} &= \frac{1}{3} \times \pi \times \left(\frac{5}{3}\right)^2 \times \frac{5\sqrt{35}}{3} \\ &= \frac{125\sqrt{35}\pi}{81} \text{ cm}^3 \end{aligned}$$

b) In $\triangle BPO$;

$$\begin{aligned} PB^2 &= 5^2 + 10^2 - 2(5)(10)(\cos 60^\circ) \\ &= 125 - 100 \times \frac{1}{2} \\ &= 75 \end{aligned}$$

$$\begin{aligned} PB &= \sqrt{75} \quad \text{PB is a length so } PB > 0, PB \neq -\sqrt{75} \\ &= 5\sqrt{3} \end{aligned}$$



$\therefore$ Length of string, $PB = 5\sqrt{3} \text{ cm}$.

1 - correct set up

2 - correct set up and acute solⁿ

3 - All correct

Q 14

a) $\sin X - 3 \cos X = 3$

$$\sqrt{10} \sin(X - \theta) = 3 \quad ; \tan \theta = 3 \Rightarrow \theta \approx 71.56^\circ$$

$$\sin(X - \theta) = \frac{3}{\sqrt{10}} \quad \& \quad -\theta \leq X - \theta \leq 360^\circ - \theta$$

$$X - \theta = 71.56^\circ \dots \text{ OR } X - \theta = 180^\circ - 71.56^\circ \dots$$

$$X = 143.13^\circ \dots \text{ OR } X = 180^\circ$$

$$\therefore X = 143^\circ \text{ OR } 180^\circ \text{ (nearest degree)} \quad \textcircled{3}$$

b) $x = vt \cos \theta$

$$y = vt \sin \theta - \frac{1}{2}gt^2$$

i) Since $PM = OM$, at P $x = -y$:

$$vt \cos \theta = \frac{1}{2}gt^2 - vt \sin \theta$$

$$\frac{1}{2}gt^2 = vt(\cos \theta + \sin \theta)$$

$$t = 0 \quad \text{OR} \quad \frac{1}{2}gt = v(\sin \theta + \cos \theta)$$

However at P, $t \neq 0$;

$$\frac{1}{2}gt = v(\sin \theta + \cos \theta) \quad \textcircled{1}$$

r/w

$$t = \frac{2v(\sin \theta + \cos \theta)}{g}$$

ii) When $t = \frac{2v(\sin \theta + \cos \theta)}{g}$,

$$x = \frac{v \cos \theta \cdot 2v(\sin \theta + \cos \theta)}{g} \quad \textcircled{1}$$

$$= \frac{v^2}{g} (2 \sin \theta \cos \theta + 2 \cos^2 \theta)$$

$$= -\frac{v^2}{g} (\sin 2\theta + \cos 2\theta + 1) \quad ; \quad \begin{matrix} \sin 2\theta = 2 \sin \theta \cos \theta \\ \text{and } \cos 2\theta = 2 \cos^2 \theta - 1 \end{matrix} \quad \textcircled{1}$$

$$\Rightarrow 2 \cos^2 \theta = \cos 2\theta + 1$$

②

iii) $OS = r$

$\Rightarrow$ When $y = 0$, $x = r$

$$vt \sin \theta - \frac{1}{2}gt^2 = 0$$

$$t(v \sin \theta - \frac{1}{2}gt) = 0$$

$$\frac{1}{2}gt = v \sin \theta \quad ; \quad t \neq 0$$

$$t = \frac{2v \sin \theta}{g} \quad \textcircled{1}$$

When $t = \frac{2v \sin \theta}{g}$,

$$x = v \cos \theta \cdot \frac{2v \sin \theta}{g} = r$$

$$\frac{2v^2 \sin \theta \cos \theta}{g} = r$$

$$\frac{v^2 \sin 2\theta}{g} = r \quad *$$

①

$$OM = \frac{4r}{3} = \frac{v^2}{g} (\sin 2\theta + \cos 2\theta + 1); \text{ from (i)}$$

$$\frac{4v^2 \sin 2\theta}{3g} = \frac{v^2 \sin 2\theta}{g} + \frac{v^2}{g} (\cos 2\theta + 1); \text{ from (i)}$$

$$\frac{v^2 \sin 2\theta}{3g} = \frac{v^2}{g} (\cos 2\theta + 1)$$

①

$$\frac{\sin 2\theta}{3} = \cos 2\theta + 1$$

Tying together.

$$\sin 2\theta = 3\cos 2\theta + 3$$

$$\sin 2\theta - 3\cos 2\theta = 3$$

③

iv) $2\theta = 143.13^\circ \text{ OR } 180^\circ$

$$\theta = 71.56^\circ \text{ OR } 90^\circ$$

①

However θ is the angle of elevation

and the projectile is not fired vertically, $0^\circ < \theta < 90^\circ$

$$\therefore \theta = 71.56^\circ$$

① Explanation

and exclusion

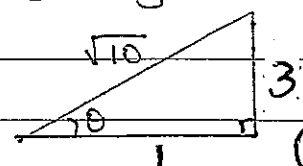
v) $x = vt \cos \theta \Rightarrow \dot{x} = v \cos \theta$

$$y = vt \sin \theta - \frac{1}{2}gt^2 \Rightarrow \dot{y} = v \sin \theta - gt$$

vi) From (a), $\tan \theta = 3$

$$\Rightarrow \sin \theta = \frac{3}{\sqrt{10}}$$

$$\& \cos \theta = \frac{1}{\sqrt{10}}$$



①

$$\dot{x} = \frac{v}{\sqrt{10}} \text{ and } \dot{y} = \frac{3v}{\sqrt{10}} - g \left(\frac{2v}{g} (\sin \theta + \cos \theta) \right) \text{ from (i)}$$

$$\dot{y} = \left(\frac{3}{\sqrt{10}} - \frac{6}{\sqrt{10}} - \frac{2}{\sqrt{10}} \right) v$$

$$= \frac{-5v}{\sqrt{10}}$$

①

$$(\text{Velocity})^2 = \dot{x}^2 + \dot{y}^2 = (kV)^2$$

$$\left(\frac{v}{\sqrt{10}} \right)^2 + \left(\frac{-5v}{\sqrt{10}} \right)^2 = k^2 v^2$$

①

$$\frac{26v^2}{10}$$

$$= k^2 v^2$$

$$k$$

$$= \sqrt{\frac{13}{5}}; \text{ since the magnitude of } kV, k > 0$$